

MONIL ARORA

Vancouver, BC

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Education

University of British Columbia, Vancouver

Expected 2027

Bachelor of Applied Science in Electrical Engineering

Relevant Coursework: Digital System Design, Computing Systems, Signals and Systems, Systems and Controls

Work Experience

Laser Zentrum Hannover (LZH)

Jan – Jul 2026

Research Intern

Hannover, Germany

- Extended **STM32F401CC** firmware in **C++/FreeRTOS** packetizing sensor telemetry into **CCSDS SPP** frames, **SLIP**-encoded over **RS-422** via **Brainboxes** serial-to-Ethernet with **COP-1/FARM** delivery and log-based **boot counter** in internal flash; diagnosed and fixed an **IWDG** reset mid image-transfer by relocating task check-ins inside the transmission loop.
- Extended **YAMCS** Mission Control System with custom **Java data links** for SLIP-over-UDP with accumulation-buffer frame reassembly, **COP-1 FOP**-compliant TC delivery, and an **XTCE mission database** decoding **APID 130 TM / 100 TC** with threshold alarms.
- Implemented end-to-end **image downlink** with an 8 KB SRAM buffer sending 79-byte **APID 131** chunks over **VCID 1** on firmware, and a Python **WebSocket** reassembler reconstructing JPEGs into **YAMCS** object storage; validated with a web-based payload simulator and **Python TCP-UDP bridge**.

Sharang Shakti Pvt Ltd

May – Jun 2024

Embedded Signal Processing Engineer

Gurugram, India

- Contributed to the **digitalization of an analog radar system** by designing and optimizing high-frequency waveform generation and signal conditioning modules using the **AD9910 DDS**, improving signal resolution and reducing phase noise.
- Integrated the **AD8336 Variable Gain Amplifier (VGA)** with the **AD9910** for real-time amplitude modulation and dynamic range adjustment in **high-frequency radar signals**.

Projects

Self-Balancing Robot | *C, Arduino, ESP, Sensors, Embedded Systems*

Jan – Apr 2025

- Designed and built a two-wheeled **self-balancing robot** using **Arduino Nano 33**, **IMU sensors**, and **motor drivers**, implementing real-time **PWM control** for tilt stabilization up to **45°**.
- Developed a **BLE-enabled mobile app** and **LCD remote** using **ESP32** to transmit commands and display RPM, tilt angle, and sonar distance readings.
- Validated proportional control through **oscilloscope analysis**, achieving approximately **30% improvement in stability**.

Deep Neural Network Accelerator | *Verilog, Assembly, C, FPGA*

Oct – Nov 2024

- Implemented a **deep neural network accelerator** on an embedded **Nios II system** using the **DE1-SoC** board for MNIST classification with **Q16.16 fixed-point arithmetic** and **ReLU** activation.
- Developed a **system-on-chip (SoC)** featuring an **SDRAM interface**, applying **Quartus timing constraints** to ensure stable operation at **50 MHz**.

Metal Detector Robot | *C, STM32, EFM8, JDY40 Bluetooth*

Mar – Apr 2024

- Collaborated with a team of six to design a battery-powered, **remote-controlled metal-detection robot** using **STM32** and **EFM8 microcontrollers**.
- Programmed robust **communication protocols** and an **LCD interface** for real-time visualization of **metal detection intensity** and **magnetic field strength (up to 1.4 T)**.

Reflow Oven Controller | *Assembly, ARM*

Jan – Feb 2024

- Developed a **microcontroller-based reflow oven controller** using the **N76E00 (ARM architecture)** in **assembly language** to automate **soldering temperature profiles** ranging from **25°C to 240°C**.
- Achieved precision within **±3°C**, integrating **automatic safety shutdowns and safety alerts**, a **real-time PC monitoring interface**, and support for multi-stage temperature ramping.

Technical Skills

Languages: C, C++, Java, Python, Assembly, Verilog, VHDL

Tools: YAMCS, FreeRTOS, PlatformIO, Maven, Quartus, ModelSim, CrossIDE, Cadence, MATLAB, LTspice, Git, VS Code

Hardware: STM32, ESP32, EFM8, Arduino, DE1-SoC, Raspberry Pi, IMU Sensors, Motor Drivers